Unit 3: Lasers & Optical Fibers

LASER is an acronym for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation.

Characteristics of Laser beam

Directionality

Monochromacity

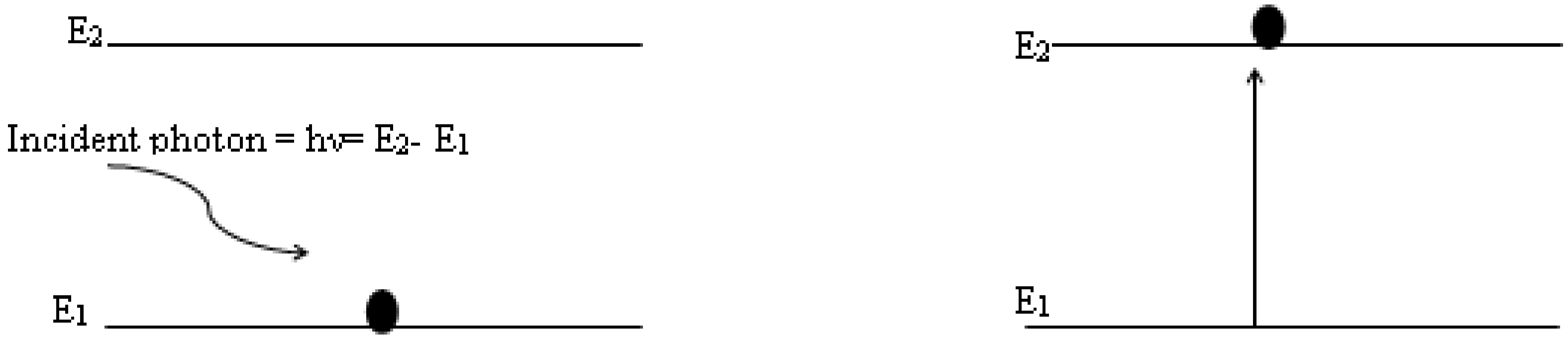
Coherence

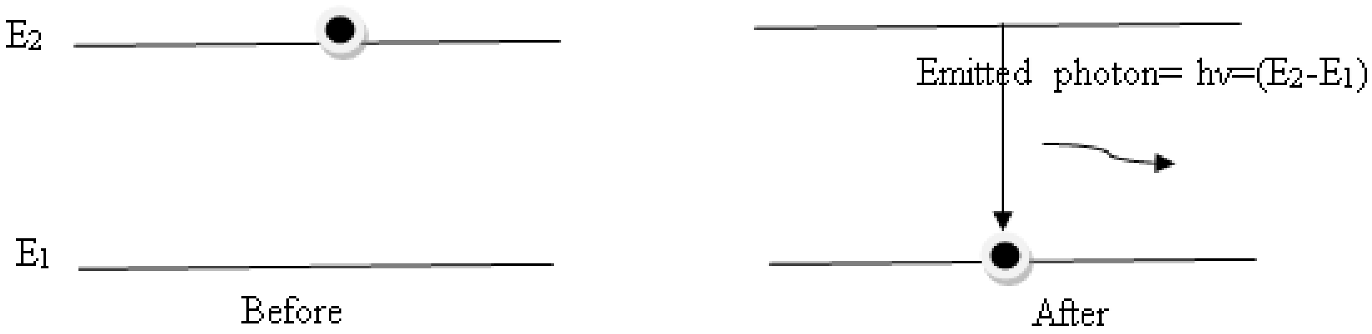
High Intensity

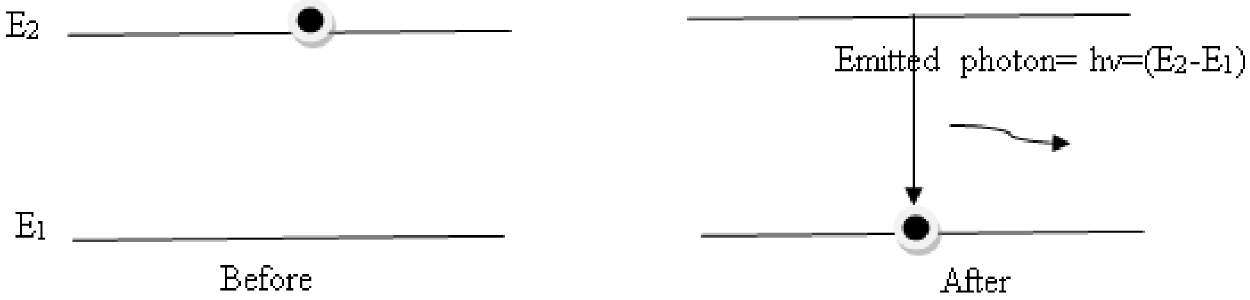
Focus ability

Basic principles: Interaction of Radiation with Matter

- Induced Absorption of radiation
- Spontaneous emission of radiation
- Stimulated emission of radiation




E2
E1
Before
Emitted photon= hv=(E2-E1)
After


E2
E1
Before
Emitted photon= hv=(E2-E1)
After

Some basic definitions

Atomic system:

It is a system of atoms or molecules having discrete energy levels.

Active medium

It is the medium in which light amplification takes place. The medium may be solid, liquid or a gas.

Energy density

The energy density U_ν refers to the total energy in the radiation field per unit volume per unit frequency due to photons. It is given by the Plank's distribution law

- $$U_\nu = \frac{8\pi h\nu^3}{c^3}\left[\frac{1}{e^{\frac{h\nu}{kT}}-1}\right]$$

Population

It is the number density (the number of atoms per unit volume) of atoms in a given energy state.

Boltzmann factor

It is the ratio between the population of atoms in the higher energy state to the lower energy state under thermal equilibrium.

$$\frac{N_2}{N_1} = e^{-\frac{h\nu}{kT}}$$

Population inversion

It is the condition such that the number of atoms in the higher energy (N_2) state is greater than the number of atoms in the ground state (N_1). i.e, $N_2 > N_1$

Expression for energy density of incident radiation in terms of Einstein coefficients:

We know that

The rate of induced absorption $= B_{12}U\nu N_{1},$

The rate of spontaneous emission $= A_{21}N_2$

The rate of stimulated emission $= B_{21}U\nu N_2$

N_1 and N_2 are the number of atoms in the energy state E_1 and E_2 respectively, B_{12}, A_{21} and B_{21} are the Einstein coefficients for induced absorption, spontaneous emission and stimulated emission respectively.

At thermal equilibrium,

Rate of induced absorption = Rate of spontaneous emission + Rate of stimulated emission

$$B_{12}N_1U_\gamma = A_{21}N_2 + B_{21}N_2U_\gamma$$

or

$$U_\gamma (B_{12}N_1 - B_{21}N_2) = A_{21}N_2$$

$$U\gamma = \frac{A_{21}N_2}{B_{12}N_1 - B_{21}N_2}$$

$$U\gamma = \frac{A_{21}}{B_{21}}\left[\frac{1}{\frac{B_{12}N_1}{B_{21}N_2} - 1}\right] \quad \text{---------} \quad (1)$$

In a state of thermal equilibrium, the populations of energy levels E_2 and E_1 are fixed by the Boltzmann factor. The population ratio is given by,

$$\frac{N2}{N1} = e^{-(\frac{E2-E1}{kT})}$$

$$N_2 = N_1 e^{-\left(\frac{E_2-E_1}{kT}\right)} = N_1 e^{-\frac{h\gamma}{kT}}$$

$$Since \ \left(E_2 - E_1 = h\nu\right)$$

$$\therefore \qquad \frac{N_1}{N_2} = e^{\frac{h\gamma}{kT}}$$

Equation (1) becomes,

$$U\gamma = \frac{A_{21}}{B_{21}} \left[\frac{1}{\frac{B_{12}}{B_{21}} e^{\frac{h\gamma}{kT}} - 1} \right] \qquad \text{------ (2)}$$

According to Planck's law of black body radiation, the equation for Uγ is,

$$U_\nu = \frac{8\pi h\nu^3}{c^3}\left[\frac{1}{e^{\frac{h\nu}{kT}}-1}\right] \text{-------}(3)$$

Now comparing the equation (2) and (3) term by term on the basis of positional identity we have

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\gamma^3}{c^3} \text{ and } \frac{B_{12}}{B_{21}} = 1 \qquad \text{or} \qquad B_{12} = B_{21}$$

This implies that the probability of induced absorption is equal to the probability of stimulated emission. Due to this identity the subscripts could be dropped, and A_{21} and B_{21} can be simply represented as A and B and equation (3) can be rewritten

$$U\gamma = \frac{A}{B}\frac{1}{[e^{\frac{h\gamma}{kT}}-1]}$$

Conditions for light amplification:-

Conditions for laser emission can be studied by taking the ratios of rate of stimulated emission to spontaneous emission and rate of stimulated emission to absorption.

At thermal equilibrium,

$$\frac{\text{Rate of Stimulated Emission}}{\text{Rate of Spontaneous Emission}} = \frac{B_{21} N_2 U_\nu}{A_{21} N_2} = \frac{B_{21} U_\nu}{A_{21}}$$

Since, $B_{21}/A_{21} = C^3 / 8\pi h\nu^3$ = constant,

$$\frac{\text{Rate of Stimulated Emission}}{\text{Rate of Induced absorption}} = \frac{B_{21}N_2U_\nu}{B_{12}N_1U_\nu} = \frac{B_{21}N_2}{B_{12}N_1}$$

$(B_{21}/B_{12} = 1)$

The stimulated emission will be larger than the absorption only when $N_2>N_1$. If $N_2>N_1$ the stimulated emission dominates the absorption otherwise the medium will absorb the energy. This condition of $N_2>N_1$ is known as ***inverted population state or population inversion.***

Requisites of a laser system:

- ***An active medium*** to support population inversion.
- ***Pumping mechanism*** to excite the atoms to higher energy levels.
- ***Population inversion***
- ***Metastable state***
- ***An optical cavity or optical resonator.***

Active medium:

It is the material medium composed of atoms or ions or molecules in which the laser action is made to take place, which can be a solid or liquid or even a gas. In this, only a few atoms of the medium (of a particular species) are responsible for stimulated emission. They are called active centers and the remaining medium simply supports the active centers.

Pumping Mechanism:

The process of supplying energy to the medium with a view to transfer the atoms to higher energy state is called **pumping.**

Optical pumping:

Electric discharge:

Inelastic atom-atom collision

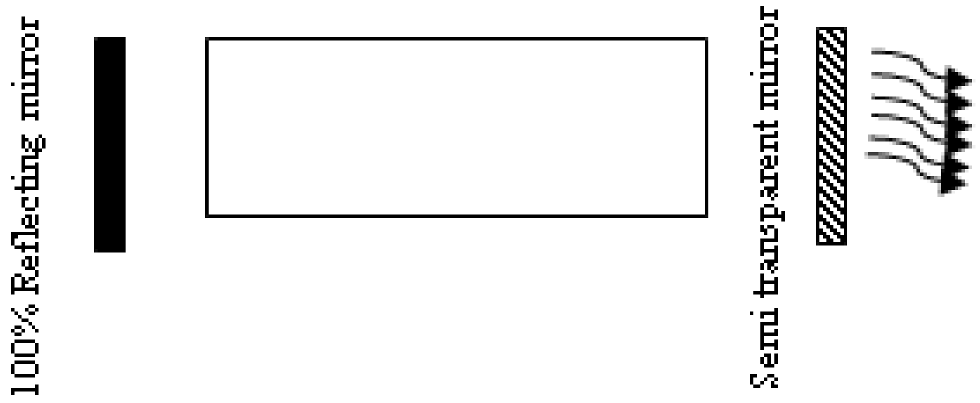
Direct conversion

Population Inversion and Meta stable state:

In order to increase stimulated emission it is essential that $N_2>N_1$ i.e., the number of atoms in the excited state must be greater than the number of atoms in the ground state. Even if the population is more in the excited state, there will be a competition between stimulated and spontaneous emission. The possibility of spontaneous emission can be reduced by using intermediate state where the life time of atom will be little longer **(10^{-6} to 10^{-3}s)** compared to excited state (10^{-9} s). This intermediate state is called **metastable state** and it depends upon the nature of atomic species used in the active medium.

Optical resonator:

The distance between the mirrors is an important parameter as it chooses the wavelength of the photons.



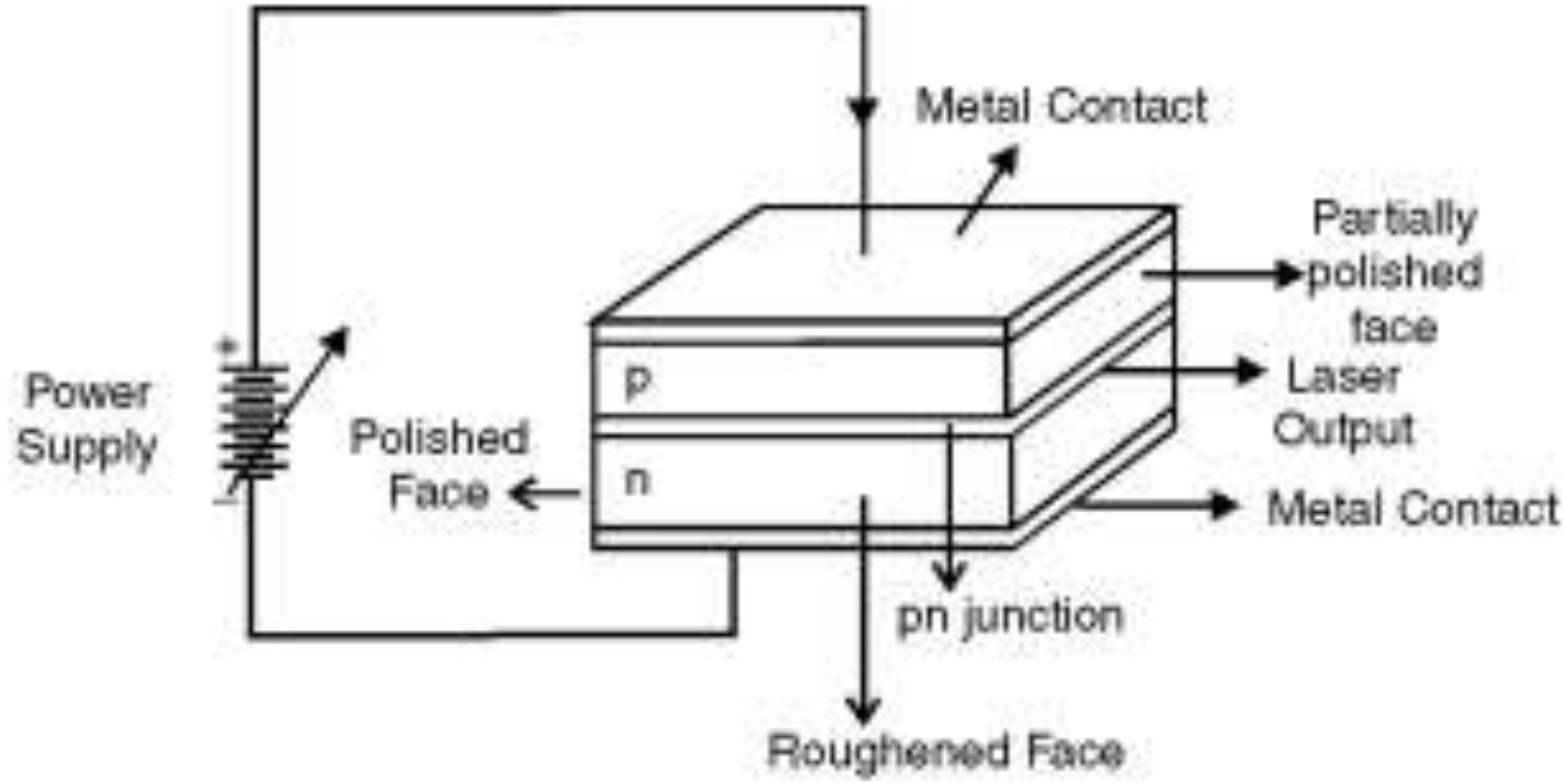
Suppose a photon is travelling between two reflectors, it undergoes reflection at the mirror kept at the other end .the reflected wave superposes on the incident wave

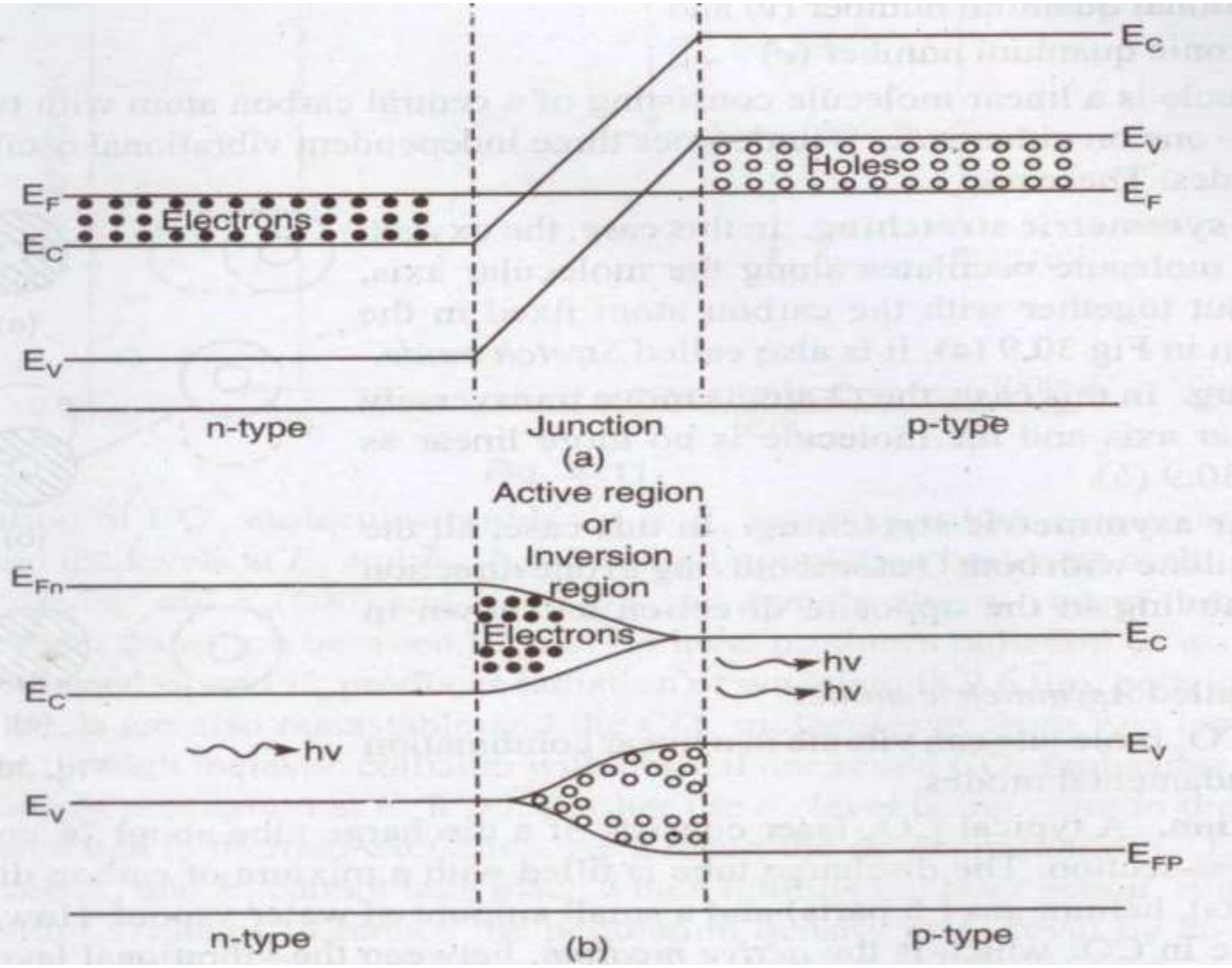
$$L = n\frac{\lambda}{2}$$

The main Role of the optical resonator is to

- Provide positive feedback of photons into the active medium to sustain stimulated emission and hence laser acts as a generator of light.
- Select the direction of stimulated photons which are travelling parallel to the axis of optical resonator and normal to the plane of mirrors are to be amplified. Hence laser light is highly directional.
- Builds up the photon density (U_ν) to a very high value through repeated reflections of photons by mirrors and confines them within the active medium.
- Selects and amplifies only certain frequencies of stimulated photons which are to be highly monochromatic and gives out the laser light through the partial reflector after satisfying threshold condition.

Semiconductor diode Laser





E_C
E_V
Holes
E_F
E_F
Electrons
E_C
E_V
n-type
Junction
p-type
(a)
Active region
or
Inversion
region
E_{Fn}
Electrons
E_C
hv
hv
E_C
hv
E_V
E_V
E_{FP}
n-type
(b)
p-type

Advantages:

1. It is simple, compact and easy to fabricate with high efficiency.
2. The laser output can be controlled by the junction current.
3. It needs lesser power than ruby and CO_2 laser.
4. It can have both continuous or pulsed output.

Disadvantages:

1. The output beam angle has a large divergence from 5^o to 15^o .
2. The mono-chromaticity is poorer than other types of laser
3. Threshold current density is very large (400A/mm2).
4. It has poor coherence and stability with low spectral purity.

Optical Attenuators as single photon sources

- Optical attenuators are used to attenuate a light beam (reduce its optical power). The amount of attenuation is specified in terms of decibels.

- If the photon flux is reduced dramatically to only 1 photon per time interval, it behaves as a **Single Photon Source.**

Types of Optical Attenuators based on attenuation mechanism

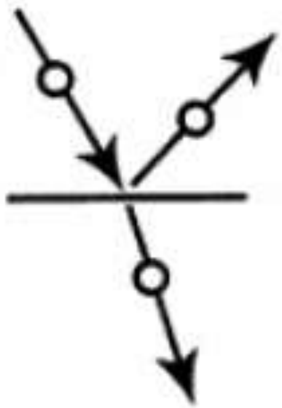
Attenuators based on Optical absorption

- Optical absorption is seen in doped glasses (absorbing filters/plates), where the type and concentration of the dopant can be used to control the amount of absorption in a certain wavelength range.
- They are low-cost, occupy less volume and have low polarization dependence.
- The removed optical power is converted into heat, hence thermal effects cause distortions of the beam profile or damage of the attenuator at high power.

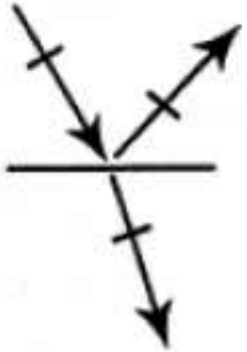
Attenuators based on Fresnel Reflection

- Fresnel reflection describes the reflection of light at the interface of two materials. The amount of light reflected depends on the angle of incidence and the polarization of the light, which defines the orientation of the electric field vector in relation to the plane of incidence.

- S-polarized light has its electric field perpendicular to the plane of incidence, while p-polarized light has its electric field parallel to the plane of incidence.



Perpendicular (S) polarization sticks up out of the plane of incidence.



Parallel (P) polarization is parallel to the plane of incidence.

- For S-polarized light, the reflection and transmission coefficient (r,t) are given by:

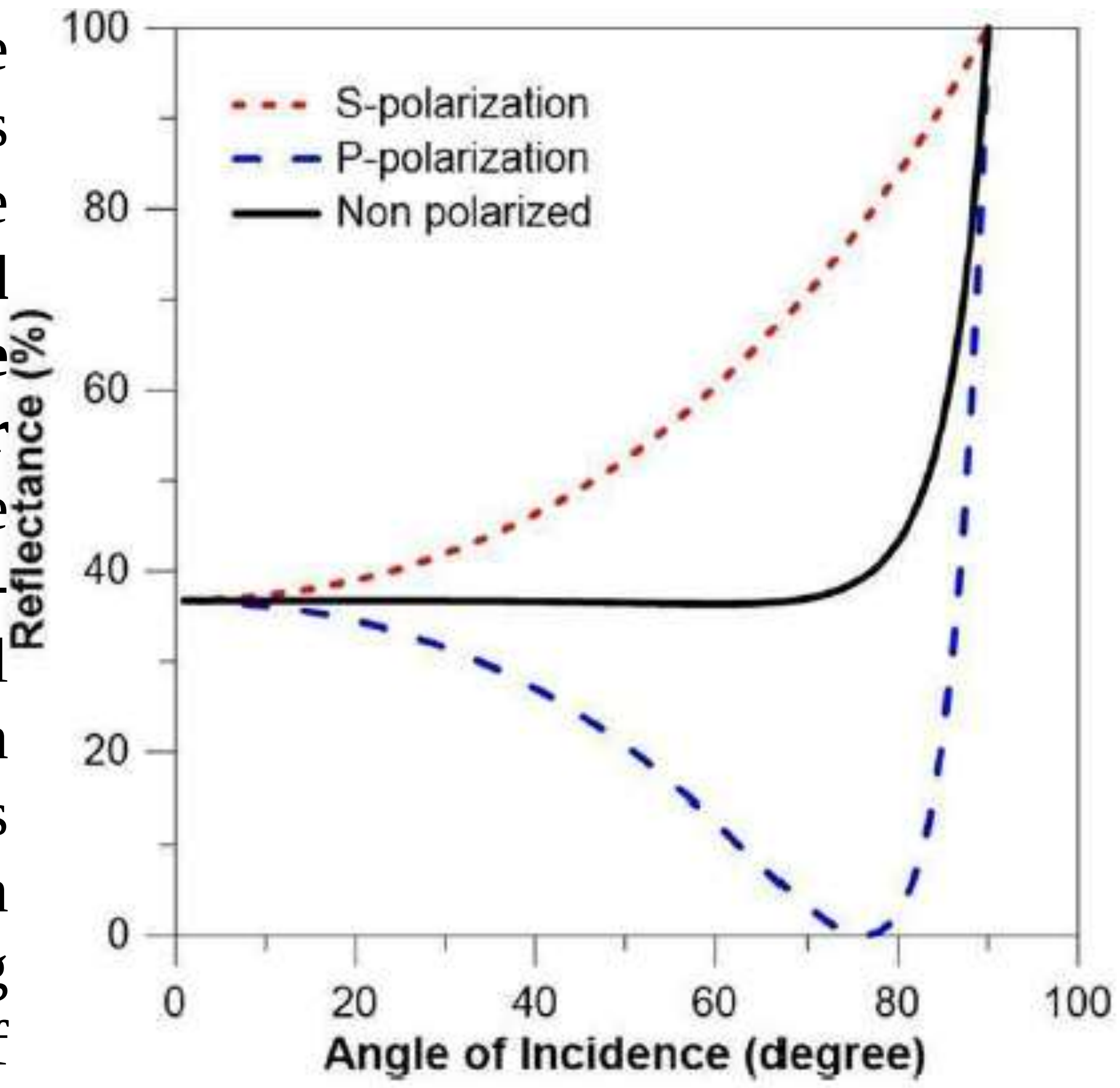
$$r = \frac{n_1 \cos\theta_i - n_2 \cos\theta_t}{n_1 \cos\theta_i + n_2 \cos\theta_t} \qquad t = \frac{2n_1 \cos\theta_i}{n_1 \cos\theta_i + n_2 \cos\theta_t}$$

For P-polarized light, the equations are given by

[No Title]

$$\frac{n_2 \cos\theta_i - n_1 \cos\theta_t}{n_2 \cos\theta_i + n_1 \cos\theta_t} \qquad t = \frac{2n_1 \cos\theta_i}{n_2 \cos\theta_i + n_1 \cos\theta_t}$$

As the incident angle reaches the **Brewster angle** given by the condition $\tan\theta_B = n_2/n_1$, the reflection co-efficient of the P-polarized light is zero, As can be seen in the figure above, the larger the angle of incidence, the higher is the difference in reflectance of s and p type polarization. **A strongly attenuated reflected P-polarized beam can be obtained with an angle of incidence near Brewster's angle, but is unsuitable,** as the presence of even a small amount of S-polarized light in the input beam will contaminate it. To remove polarization dependency of attenuation, one uses multiple reflections at small angles in sequence. Alternatively, a highly reflecting coating is used to obtain a high degree of attenuation for the **transmitted light**.



Optical Modulators

An optical modulators is a device used to control or modify the property of light, such as amplitude, phase, polarization or frequency. They are used in telecommunications, signal processing and laser systems.

Types of optical modulators

- Electro-optic modulators (EOM)
- Acoustic-optic modulators (AOM)
- Electro-absorption modulators (EAM)

Types of Optical Modulators

- EOM uses the electro-optic effect, where the refractive index of a material changes in response to an electric field. They are used fiber optic networks. There are two types in this namely **Pockel's effect** and **Kerr effect**.
- AOM are based on acoustic-optic effect in which sound waves interact with light waves within a material and causes diffraction. This leads to modulation of light. AOMs are used for beam deflection, frequency shifting and intensity modulation.
- EAM are working on the principle of electro-absorption effect. Each material has the ability to absorb light and absorption coefficient changes with the application of an electric field. As a result the light is modulated. EAMs are suitable for integrated photonic circuits.

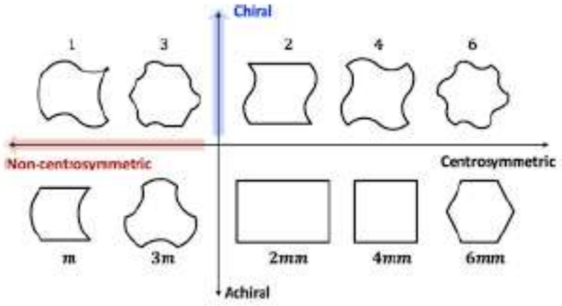
Pockel's effect (linear electro-optic effect)

When electric field is applied to some transparent crystals, the refractive index (RI) of the crystal changes. As the applied voltage increases the RI linearly. The incident light splits into two parts due to change in the RI and hence birefringence occurs. This is know as **Pockel's effect**.

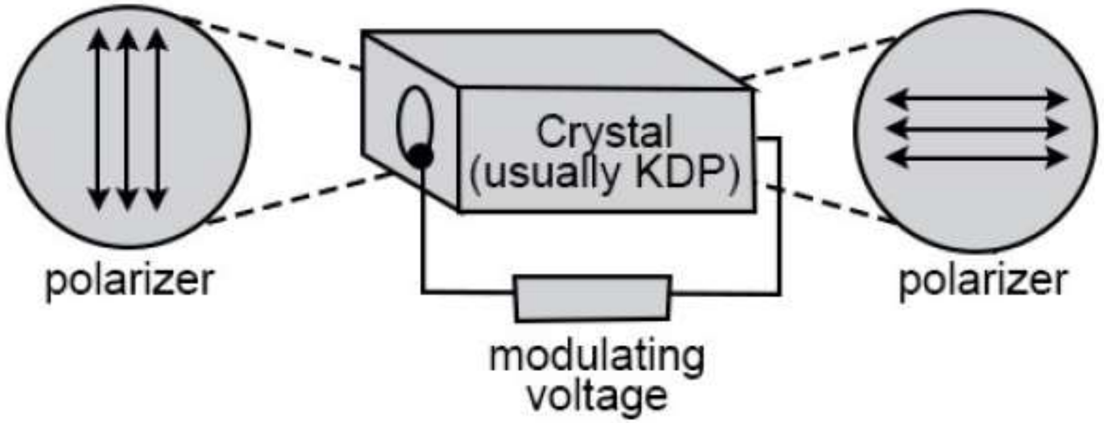
$$\Delta n \propto E$$

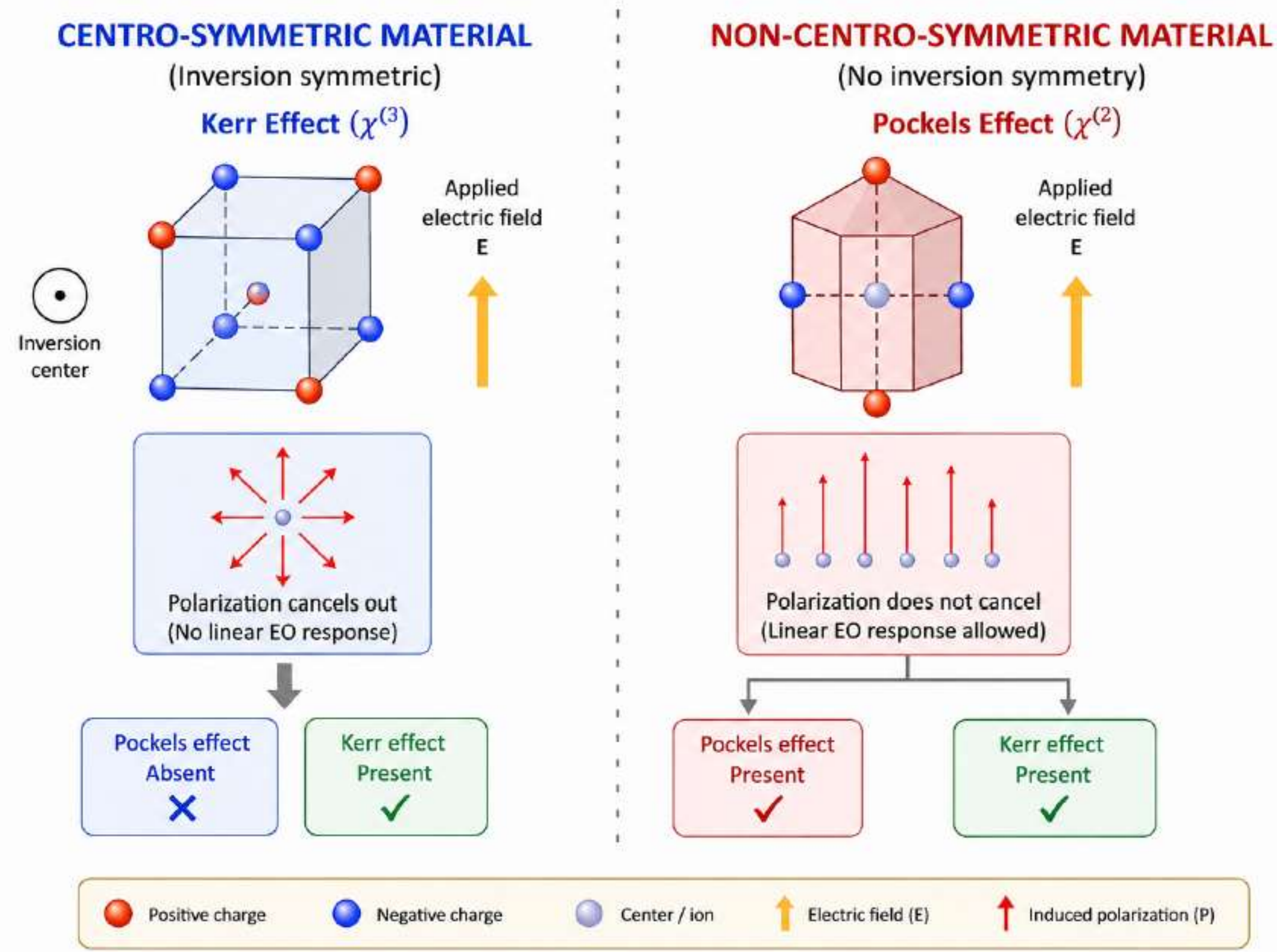
Pockel's effect occurs only in non-centro symmetric materials such as lithium niobate (LiNbO3), lithium tantalate (LiTaO3), potassium di-deutarium phosphate (KDP), β-barium borate (BBO) and compund semiconductors such as gallium arsenide (GaAs) and indium phosphide (InP).

A non-centrosymmetric structure or molecule lacks a center of inversion symmetry. If you map a point from one side of the structure, no identical point exists directly opposite through a central origin point.



Using Pockels effect the phase, polarization state and amplitude of the light passing through the crystals can be modulated by applying electric field.




CENTRO-SYMMETRIC MATERIAL
(Inversion symmetric)
Kerr Effect ($\chi^{(3)}$)
Inversion center
Applied electric field E
Polarization cancels out
(No linear EO response)
Pockels effect Absent
Kerr effect Present
NON-CENTRO-SYMMETRIC MATERIAL
(No inversion symmetry)
Pockels Effect ($\chi^{(2)}$)
Applied electric field E
Polarization does not cancel
(Linear EO response allowed)
Pockels effect Present
Kerr effect Present
Positive charge
Negative charge
Center / ion
Electric field (E)
Induced polarization (P)

Kerr Effect

When electric field is applied to transparent crystals, the refractive index(RI) of the crystal changes. The change in the RI is directly proportional to square of the applied voltage or square of the applied field strength. This is known as **Kerr effect**.

The RI is different for light polarized parallel to and perpendicular to applied field. This change makes the crystal birefringent (even though it was not birefringent earlier!). Hence the crystal used as 'electrically controllable wave plate'.

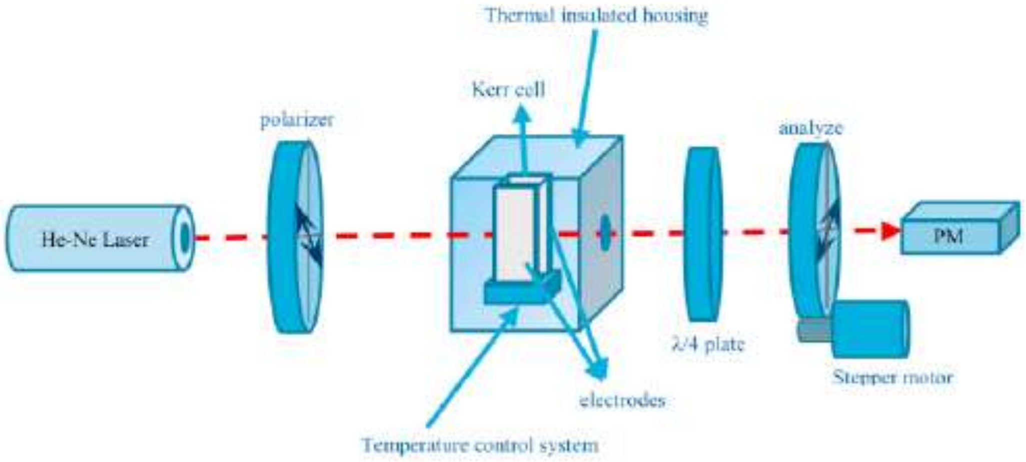
If Δn is the change in RI of the beams, λ is the wavelength of the incident light and E is the applied field strength then

$$\Delta n = \lambda K E^2$$

Where K is known as Kerr constant.

All materials show Kerr effect, but certian liquids like nitro toluene and nitro benzene display it more strongly than others.

Kerr Cell



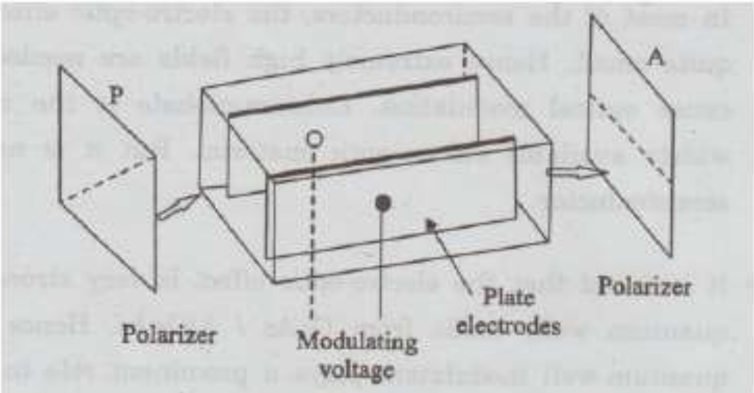


Fig. 4.30

Kerr cell is a device used to modify the polarization and amplitude of the light. It consists of a container containing liquids like nitro toluene or nitro benzene placed between the electrodes.

The Kerr cell is placed between two crossed polarizers. With no voltage applied, the polarizers block all light from passing through them. When voltage is applied, the Kerr effect causes the polarization of the light to rotate

Difference between Pockel's and Kerr effect

Feature	Pockel's Effect	Kerr Effect
Field Dependence	Linear ($\Delta n \propto E$)	Quadratic ($\Delta n \propto E^2$)
Symmetry Requirements	Non-centrosymmetric crystals	All materials
Speed/Response Time	Extremely fast (picosecond response)	Extremely fast
Applications	Pockels cells, Laser Q-switching, fiber optic modulators	Kerr cell, Optical switching, high-power laser physics

Photo Detectors

- Photodetectors (or photosensors) are optoelectronic devices that detect light or electromagnetic radiation and convert it into a proportional electrical signal (current or voltage).
- They are the fundamental components in optical receivers, digital cameras and fiber optic communication systems.

There are different types of photo detectors.

- Some of them are Single Photon Avalanche photodiode (SPAD)
- Superconducting Nanowire Single Photon Detectors (SNSPDs)

Photodiode

Photodiode is a p-n junction that converts light energy into an electric current. These diodes are designed to work in reverse bias conditions.

Working:

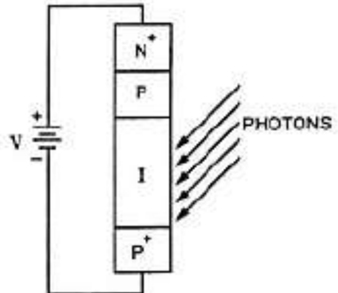
- When light is incident on p-n junction connected in reverse bias mode, it generates electron hole pair.
- The created charges separate out and move in the opposite direction and constitute electric current called photocurrent.
- By measuring the current we can detect the incident signal. Normally this current is very less and hence difficult to detect faint optical signals.
- But Avalanche Photodiode (APD) is capable of measuring a week signals also.

Avalanche Photodiode

- **APD** is a p-n junction operated at high reverse bias voltage.
- Because of higher voltage the avalanche breakdown takes place in the junction and current increases drastically due to carrier multiplication.
- This provides substantial internal amplification and hence light sensing ability increases.

Construction:

- APD consists of four layers namely heavily doped p region (P_+), intrinsic region (I), moderately doped p region as (P) and heavily doped n region (N_+). I region is relatively large in size compare to other regions.



Working:

- There exists electric field along the length of the APDs and increases towards the PN_+ region.
- The biasing voltage is applied across P_+ and N_+ region. When photon incident at P_+I junction, electron-hole pair is created.
- Holes drift towards P+ and electrons drift towards PN_+ junction.
- During their motion, electrons are accelerated due to the presence of electric field and ionize the atoms present in the I region (called Impact ionization).
- This is a cumulative effect and builds up into avalanche. Hence the carrier multiplication takes place which leads to internal amplification.
- As a result the responsivity of the device increases. APDs can be operated even at a very low intensities of light.

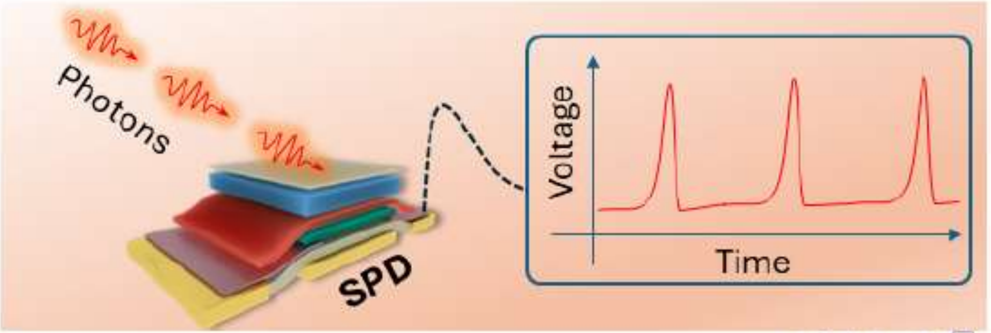
- It has to be 'quenched' (reduced). This is done by two ways

1. By using suitable resistor which limits the current (called passive quenching)
2. By using suitable electronic circuit which quickly lowers the voltage by detecting rise in the current (active quenching) and restores it after 'dead time'.

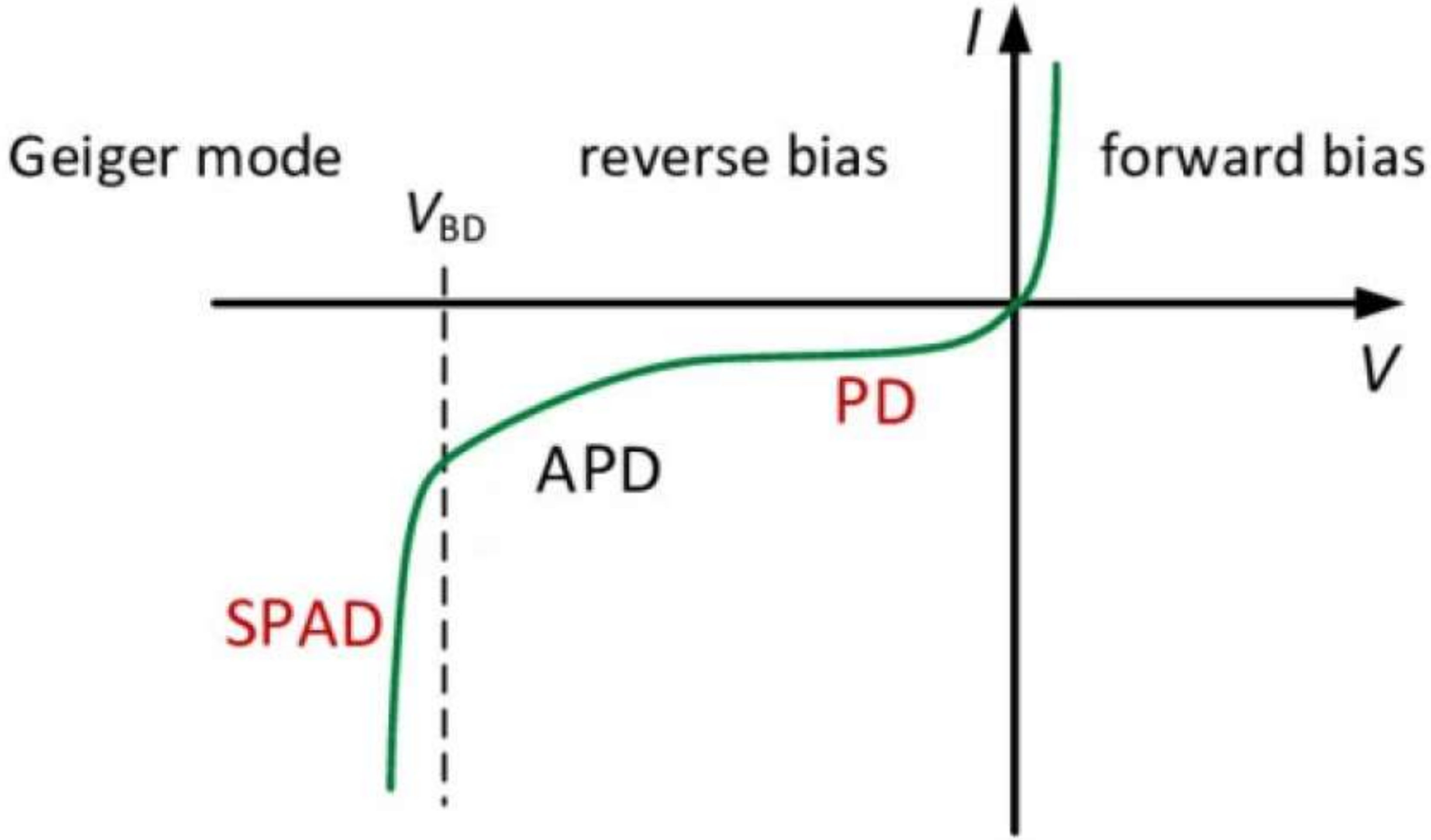
As a result a sharp digital like pulse is obtained as shown in fig for every single photon incident on the SPAD.

This is how single photon is detected by the diode.

SPADs are used in fluorescence spectroscopy, bio photonics, Lidars, Quantum computing, QKDs etc.



In the following I-V characteristics graph the working regions of photodiode, APD and SPAD are mentioned.

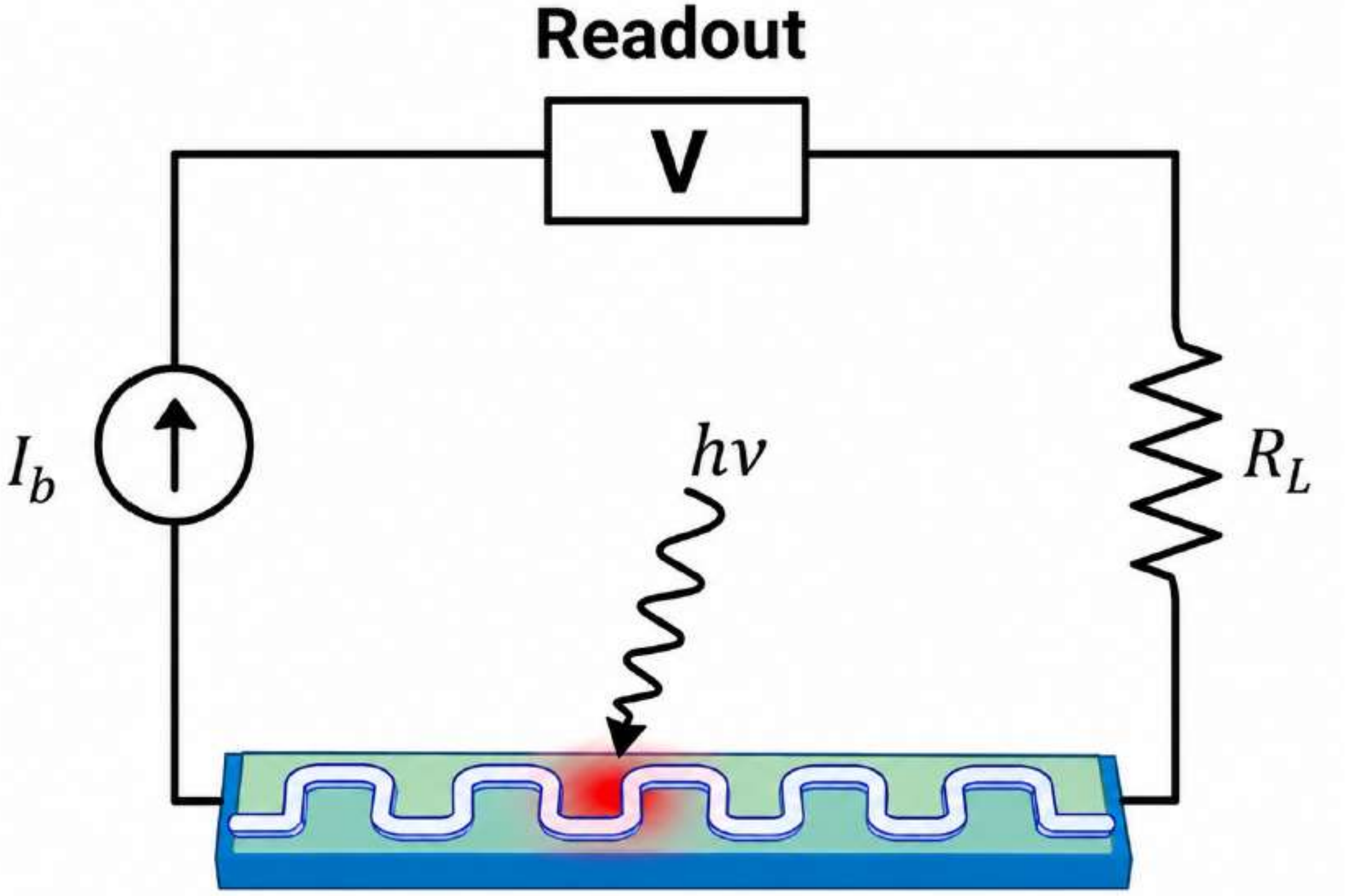


Superconducting nanowire single photon detector

- Superconducting nanowire Single Photon Detectors (SNSPDs) are the advanced devices for single-photon detection. They work on the principle that when a photon is incident it breaks Cooper pairs and reduces the local critical current. By measuring the changes in the current one can detect the photon.

Construction :

The SNSPD consists of a thin and narrow superconducting wire. The wire is arranges in a curved fashion (called Meander geometry) to achieve high detection efficiency. The wire is kept at a temperature below the critical temperature and connected to DC source. The biasing current (I_b) in the wire is maintained just below its critical current (I_c). The whole arrangement is connected to a special type of electronic circuit called 'readout'.



Readout
V
I_b
$h\nu$
R_L

Working:

- When a photon is incident on the wire, it breaks the Cooper pairs and the local critical current decreases (fig).
- If the critical current decreases below the bias current then the superconductivity is spoiled and a local non-superconducting region or 'hotspot' is created.
- The resistance of the hotspot is larger than the resistance of the readout (about $1k\Omega$). As a result bias current flows through the readout. (In fact readout acts a shunt for the current).
- This produces a measurable voltage pulse. such that 'one photon one pulse'. By measuring this voltage we can detect the incident photon.
- With most of the bias current flowing through the readout, the non-superconducting region rapidly cools and returns to the superconducting state, ready to detect another photon.

Advantages:

SNSPDs offer many important advantages over SPADs, some of them are

- Unparalleled detection efficiency and timing precision.
- Negligible intrinsic dark current.
- Ultrafast detection rates.
- Broadband operation

Uses:

They are becoming more popular in the applications of bio photonics, quantum optics and quantum computing.

Single Photon Avalanche Photodiode

- If the reverse voltage is above the breakdown voltage then APD is said to be in Geiger mode.
- In this mode APD detects single photon and hence it is called single photon avalanche phot diode (SPAD).
- The working mechanism of of SPAD is same as APD with few changes.
- The incident single photon at $P_{+}I$ junction creates hole-electron pair and the electron triggers avalanche current due to impact ionization.
- Since the reverse voltage is above the breakdown voltage, the number of electrons increase suddenly and the current shoots up rapidly.

Advantages:

SNSPDs offer many important advantages over SPADs, some of them are

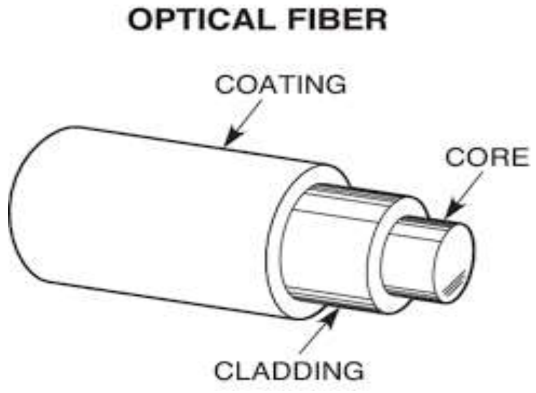
- Unparalleled detection efficiency and timing precision.
- Negligible intrinsic dark current.
- Ultrafast detection rates.
- Broadband operation

Uses:

They are becoming more popular in the applications of bio photonics, quantum optics and quantum computing.

Optical Fibers

Optical fibres are the light guides used in optical communications as wave-guides. They are thin, cylindrical, transparent flexible dielectric fibres. They are able to guide visible and infrared light over long distances.

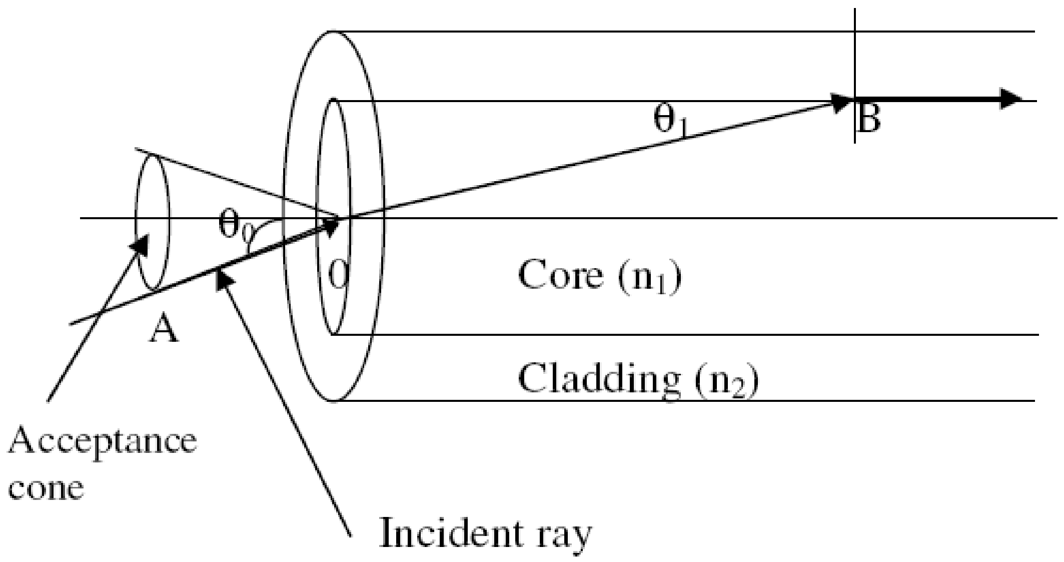


Propagation mechanism in Optical fibre:

In optical fibres light waves can be guided through it, hence are called light guides. The cladding in an optical fibre always has a lower refractive index (RI) than that of the core. The light signal which enters into the core can strike the interface of the core and the cladding at angles greater than critical angle of incidence because of the ray geometry. The light signal undergoes multiple total internal reflections within the fibre core. Since each reflection is a total internal reflection, the signal sustains its strength and also confines itself completely within the core during propagation. Thus, the optical fibre functions as a wave guide.

Numerical Aperture and Ray Propagation in the Fibre:

Let n_0, n_1 and n_2 be the refractive indices of surrounding air medium, core and cladding material respectively. The RI of cladding is always lesser than that of core material ($n_2 < n_1$) so that the light rays propagate through the fibre.



The ray travels along AO entering into the core at an angle of θ_0 with respect to the fibre axis. Let it be refracted along OB at an angle θ_1 in the core and further proceed to fall at critical angle of incidence (= 90-θ_1) at B on the interface of core and cladding. Since it is critical angle of incidence, the refracted ray grazes along core and cladding interface.

It is clear from the figure that a ray that enters at an angle of incidence less than θ_0 at O, will have to be incident at an angle greater than the critical angle at the core-cladding interface, and gets total internal reflection in the core material.

Sin θ_0 is called the numerical aperture (N.A.) of the fibre. It determines the light gathering ability of the fibre and purely depends on the refractive indices of core, cladding and surrounding medium.

By applying the Snell's law at O,

$$n_0 sin\theta_0 = n_1 sin\theta_1$$

$$sin\theta_0 = \frac{n_1}{n_0} sin\theta_1 \text{------------}(1)$$

At the point B on the core and cladding interface, the angle of incidence = 90-θ_1

Applying Snell's law at B

$$n_1 \sin(90 - \theta_1) = \quad n_2 sin90$$

or $\quad n_1 cos\theta_1 = n_2$

$$cos\theta_1 = \frac{n_2}{n_1} \text{-------------------}(2)$$

From equation (1)

$$sin\theta_0 = \frac{n_1}{n_0} sin\theta_1 = \frac{n_1}{n_0}(\sqrt{1 - cos^2\theta_1})$$

$$sin\theta_0 = \frac{n_1}{n_0}\sqrt{1 - \frac{n_2^2}{n_1^2}} = \frac{n_1}{n_0} x \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}} = \frac{\sqrt{n_1^2 - n_2^2}}{n_0}$$

If the medium surrounding the fibre is air, then $n_0 = 1$,

Therefore, $sin\theta_0 = \sqrt{n_1^2 - n_2^2}$

$\sin\theta_0$ = Numerical aperture: NA

Therefore, $NA = sin\theta_0 = \sqrt{n_1^2 - n_2^2}$

If θ_i is the angle of incidence of an incident ray, then the ray will be able to propagate,

If $\theta_{i<}\theta_0$

or if $\sin\theta_i < \sin\theta_0$

- or $\sin\theta_i < NA$

Fraction Index Change (Δ):

The fractional index change Δ is the ratio of the difference in the refractive indices between the core and the cladding to the refractive index of core of an optical fibre. It is also known as relative core clad index difference, denoted by Δ.

If n_1 and $n_2$are the refractive indices of core and cladding, then,

$$\Delta = \frac{(n_1 - n_2)}{n_1}$$

Relation between NA and Δ

We have RI change $\Delta = \frac{n_1 - n_2}{n_1}$

Or $\Delta n_1 = (n_1 - n_2)$-----------(1)

We know that,

$$N.A. = \sqrt{n_1^2 - n_2^2}$$

$$\sqrt{(n_1 + n_2)(n_1 - n_2)}$$

$$\sqrt{(n_1 + n_2)n_1\Delta}$$

Since $n_1 \sim n_2$, $(n_1 + n_2) = 2n_1$

$$N.A. = \sqrt{2n_1^2\Delta}$$

$$N.A. = n_1\sqrt{2\Delta}$$

The value of NA can be increased by increasing the value of Δ, so as to receive maximum light into the fibre. However, fibres with large Δ will not be useful for optical communication due to the occurrence of a phenomenon inside the fibre called multipath dispersion or intermodal dispersion. This phenomenon introduces a time delay factor, in the travel length and may cause distortion of the transmitted optical signal.

Modes of Propagation:

The possible number of paths of light in an optical fibre determines the number of modes available in it. It also determines the number of independent paths for light that a fibre can support for its propagation without interference and mixing.

Single mode fiber

Multi mode Fiber

The number of modes supported for propagation in the fibre is determined by a parameter called V-number. If the surrounding medium is air then the V- number is given by

$$V = \frac{\pi d}{\lambda}\sqrt{(n_1^2 - n_2^2)}$$

For step index fibre, the number of modes $\approx \frac{V^2}{2}$

For graded index fibre, the number of modes $\approx V^2/4$

Step index fibre	Graded index fibre
• Refractive index of core is uniform • Propagation of light is in the form of meridional rays • Step index fibres has lower bandwidth • Distortion is more (in multimode) • No. of modes for propagation $N_{step} = V^2/2$	• Refractive index of core is not uniform • Propagation of light is in the form of skew rays • Graded index fibres has higher bandwidth • Distortion is less • No. of modes for propagation $N_{grad} = V^2/4$

Types of Optical fibres:

The optical fibres are classified under 3 categories. They are

a) Step index Single mode fibre (SMF)

b) Step index multi modefibre (MMF)

c) Graded index Multi Mode Fibre (GRIN)

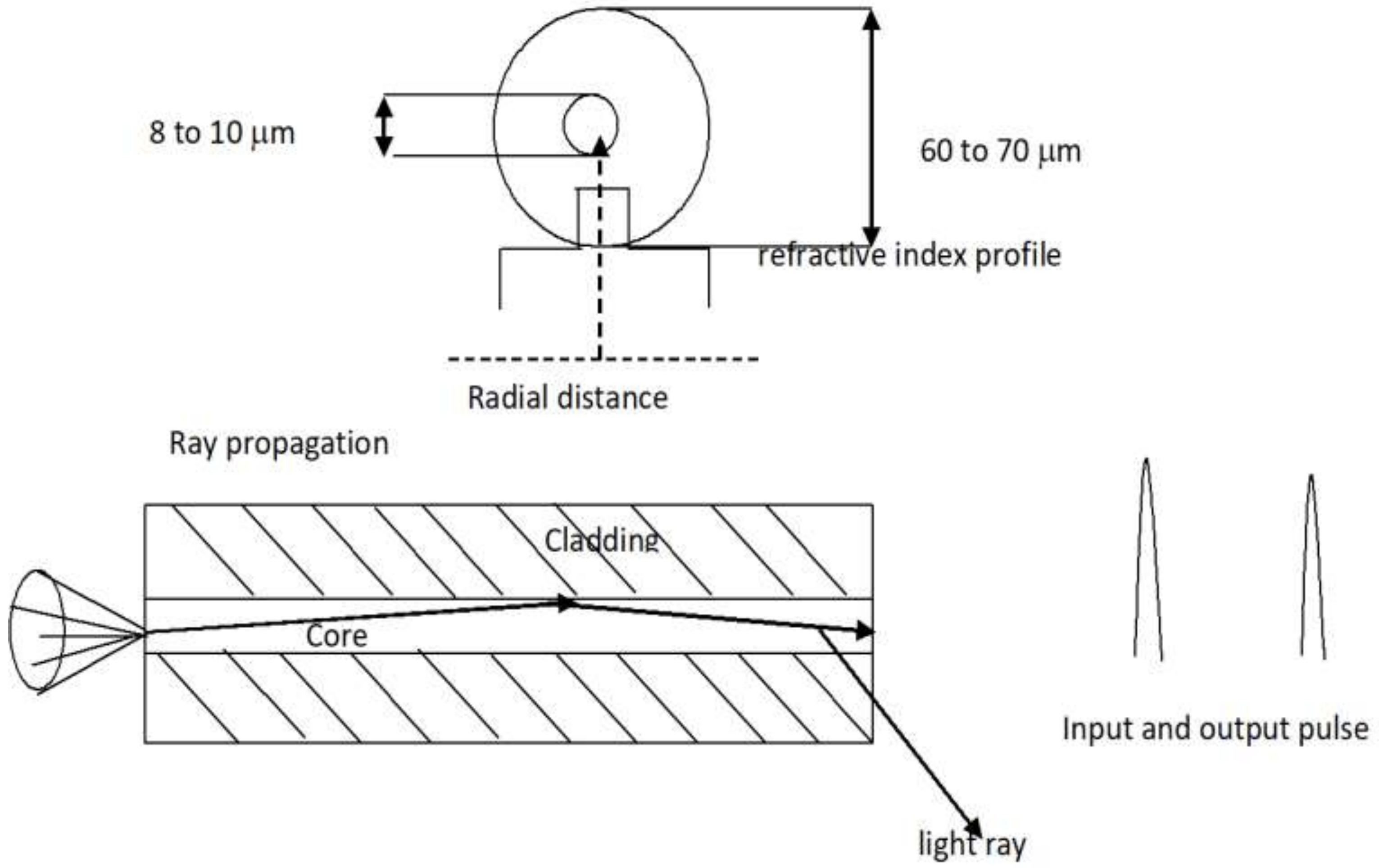


Fig: Step Index Single Mode Fibre

Since single mode is propagating through the fibre, intermodal dispersion is zero in this fibre. They find particular application in submarine cable system.

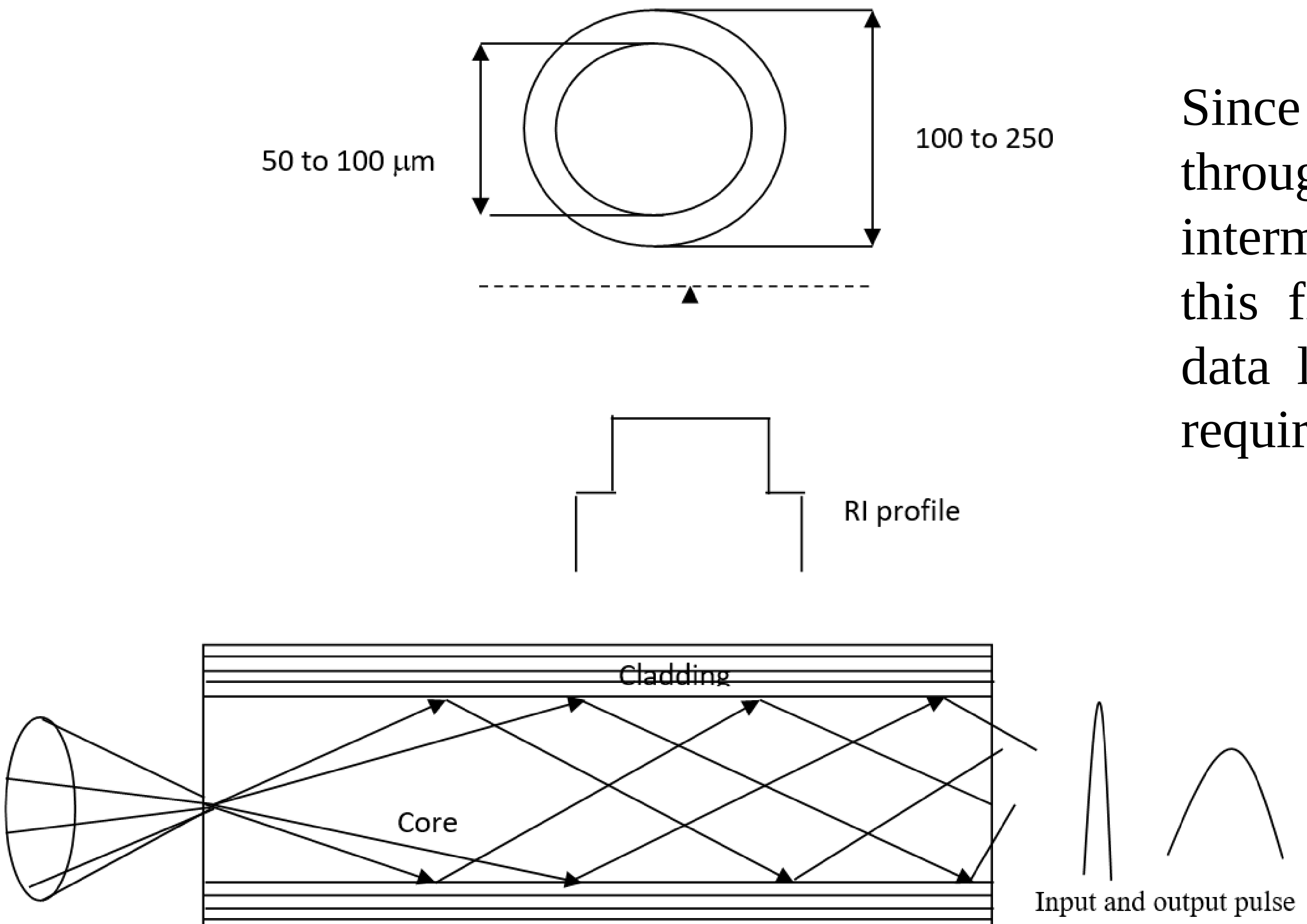


Fig. Step index multimode fibre

Since multi modes are propagating through this fibre with different paths, intermodal dispersion is maximum in this fibre. Its typical application is in data links which has lower bandwidth requirements.

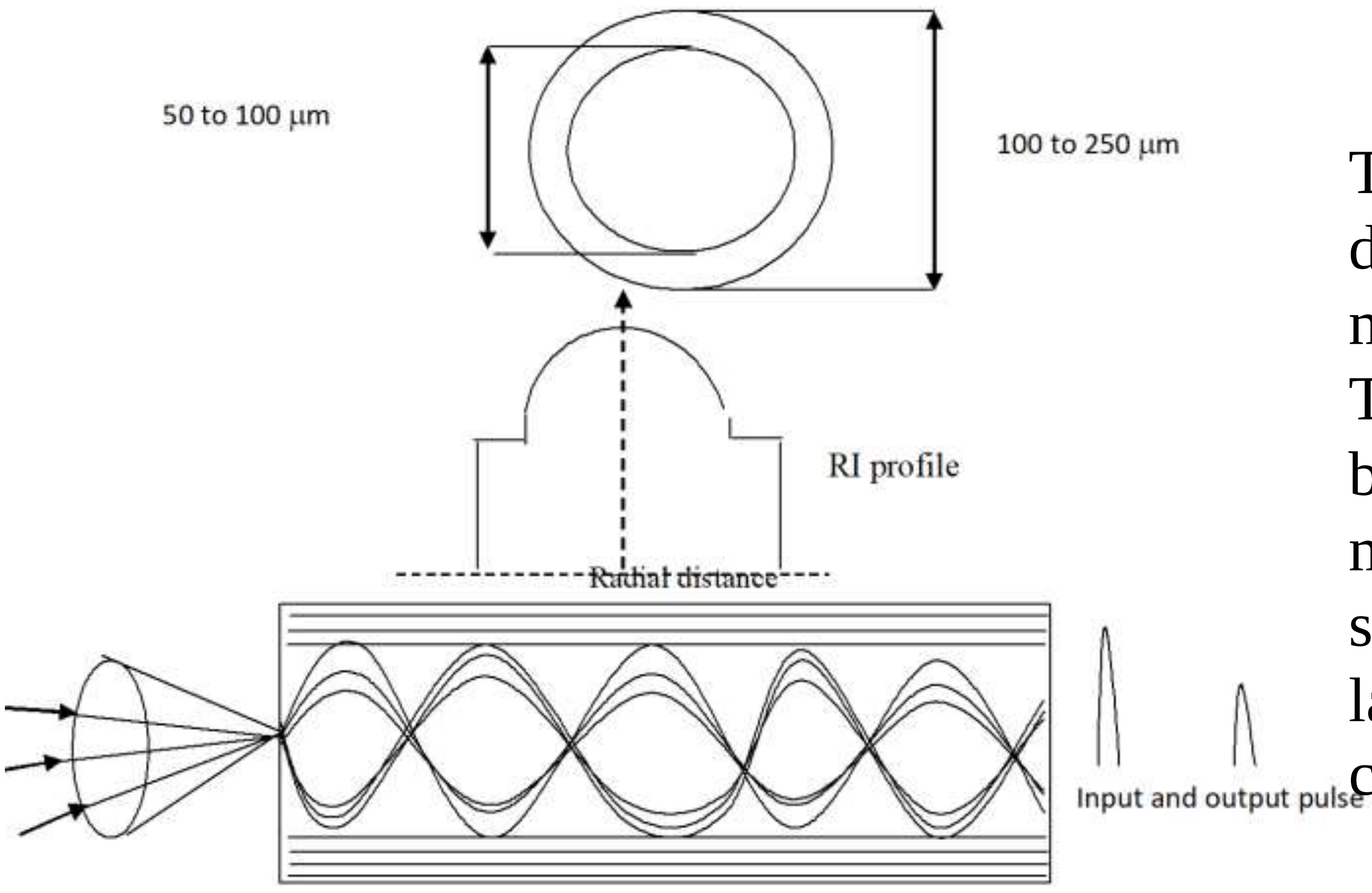


Fig. Graded index multimode fibre

This reduces the effect of intermodal dispersion and hence losses are minimum with little pulse broadening. These fibres are most suitable for large bandwidth, medium distance and medium bit rate communication systems. For such cables, either a laser or LED source can be used to couple the signal into the core.

Attenuation in optical fibres:

The total energy loss suffered by the signal due to the transmission of light in the fibre is called **attenuation.**

The important factors contributing to the attenuation in optical fibre are i) Absorption loss ii) Scattering loss iii) Bending loss

iv) Intermodal dispersion loss and v) coupling loss.

Attenuation is measured in terms of attenuation co-efficient and it is the loss per unit length. It is denoted by symbol α. Mathematically attenuation of the fibre is given by,

$$\alpha = -\frac{10 \times \log_{10}\left(\frac{P_{out}}{P_{In}}\right)}{L} \text{ dB/km}$$

Where P_{out} and P_{in} are the power output and power input respectively, and L is the length of the fibre in km.

Therefore, Loss in the optical fibre = $\alpha \times L$